

# EXAM 1

## STEP 1: SCHEDULE YOUR SESSION

Hyperlink: [Ordinary Differential Equation Exam](#)

## STEP 2: WORK ON YOUR EXAM

THE EXAM HAS TWO SETS OF PROBLEMS, X and Y  
CHOOSE ONLY ONE SET AND FOLLOW THE PROMPTS

## STEP 3: SUBMIT AND SHOW UP

MAKE SURE THAT YOU FOLLOWED THE GUIDELINES

### Guidelines<sup>1</sup>

#### Written Part

- Your work must be submitted before your scheduled session, or handed in at the start of the session.
- Your scheduled session will not proceed if your written work has not been submitted or is incomplete.

#### Oral Part

- You must sign-up for a 15-minute time slot on exam days at least 24 hours in advance.
- If you had discussed an AES accommodation, you may schedule two consecutive sessions.
- You must show up to your scheduled oral part of the exam in-person and on time.
- A no-show or being late results in an automatic “M” (missed exam).

#### Submission

- Submit your work in Moodle before your scheduled session.
- If necessary, you may hand in a physical copy at the start of the session.

#### Collaboration and Academic Integrity Policy

- You may collaborate with your peers, but the work you submit must reflect your own.
- You may use all resources available to you, that includes your notes, books, and online resources.
- Be prepared to defend all aspects of your work.

#### Extra Credit Opportunity

- Optional problems will count as an extra credit added to your homework total.

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<sup>1</sup>Please review the course Syllabus for the complete Exam guidelines and assessment process.

## X. Population Growth in a City

### Description:

The city's population (scaled in 10 thousand) is represented by the function  $P(t)$ , where  $t \geq 0$  is the time in years. The model for this city's population growth uses the following assumptions:

- The birth rate is  $\frac{1}{10}$ . That is 1 thousand people per year.
- The immigration/emigration rate is  $-\infty < \beta < \infty$ .
- The carrying capacity is 100, that is 1 million people that the city can sustain.
- The city's initial population is  $P_0$ .

The population change over time is governed by the following ODE:

$$P' = \frac{1}{10}P(100 - P) + \beta.$$

### Written Part:

1. Classify the given ODE in terms of order, linearity, autonomy, and homogeneity as you change  $\beta$ .
2. Show that the *equilibrium solutions* are

$$P_1 = 50 - \sqrt{10}\sqrt{250 + \beta} \quad \text{and} \quad P_2 = 50 + \sqrt{10}\sqrt{250 + \beta}.$$

3. Draw complete and well-represented *phase diagrams* for cases  $\beta < -250$ ,  $\beta = -250$ , and  $\beta > -250$ .
4. Draw a complete and well-represented *slope field* for the two-equilibrium case. Then, draw at least four solution curves on the slope field.
5. Assume real numbered  $P_1 \neq P_2$ . Use the *separation of variables* method to show that the *general solution* is

$$P(t) = \frac{-AP_2e^{(P_1-P_2)t} + P_1}{1 - Ae^{(P_1-P_2)t}},$$

where  $A$  is the integration constant.

6. Using the initial condition  $P(0) = P_0$ , show that the integration constant  $A$  for the *specific solution* is

$$A = \frac{P_0 - P_1}{P_0 - P_2}.$$

### Oral Part:

7. Describe the meaning of  $\frac{1}{10}P$ ,  $(100 - P)$ , and  $\beta$  in context of the problem description.
8. Explain the key steps of the method of separation of variables.
9. For which values of initial conditions  $P_0$  does the solution reach a zero population, and for which values does it not? Consider all three cases  $\beta < -250$ ,  $\beta = -250$ , and  $\beta > -250$ .

### Optional:

10. Use Euler's method with step size of 0.10 in 100 steps to approximate the specific solution of the ODE, then compare your approximations to the exact specific solution.

## Y. Salt Solution in a Fluid Tank

### Description:

The mass of salt in a tank is represented as a function  $C(t)$ , where  $t \geq 0$  is in hours. The model for the salt amount uses the following assumptions:

- The tank initially contains  $C_0$  kilograms of salt dissolved in 1000 liters of water.
- Water that contains  $\frac{1}{2}$  kilogram of salt per liter is poured at the rate of 10 liters per hour.
- The volume changes in time at a rate of  $-10 < M < 10$  liters per hour.

The change of the salt amount in the solution can be described by the following ODE:

$$C' = 5 - \frac{(10 - M)}{1000 + Mt} C.$$

### Written Part:

1. Classify the given ODE in terms of order, linearity, autonomy, and homogeneity as you change  $M$ .
2. Show that the *nullcline* is

$$C(t) = \frac{5000}{10 - M} + \frac{5M}{10 - M} t.$$

3. Draw complete and well-represented *slope fields* for cases  $-10 < M < 0$ ,  $M = 0$  and  $0 < M < 10$ . Then, draw at least four solution curves on each slope field.
4. Draw a complete and well-represented *phase diagram* for the autonomous case.
5. Use the *integrating factors* method to show that the *general solution* is

$$C(t) = 500 + \frac{M}{2}t + A(1000 + Mt)^{\frac{M-10}{M}}$$

where  $A$  is the integration constant.

6. Using the initial condition  $C(0) = C_0$ , show that the integration constant  $A$  for the *specific solution* is

$$A = \frac{C_0 - 500}{1000^{\frac{M-10}{M}}}.$$

### Oral Part:

7. Describe the meaning of ODE terms 5 and  $-\frac{(10 - M)}{1000 + Mt}C$  in context of the problem description.
8. Explain the key steps of the method of integrating factors.
9. For which values of initial conditions  $C_0$  does the solution attain a critical point (minimum/maximum), and for which values of  $C_0$  does it not? Consider all three cases  $-10 < M < 0$ ,  $M = 0$  and  $0 < M < 10$ .

### Optional:

10. Use Euler's method with step size of 0.10 in 100 steps to approximate the specific solution of the ODE, then compare your approximations to the exact specific solution.